

HEALTHCARE ANALYTICS

Definition

Healthcare analytics leverages clinical and administrative data to generate actionable insights for better decision-making. This process optimizes patient outcomes and operational efficiency across the entire healthcare industry.

4 types of healthcare analytics

• Predictive analytics

It uses machine learning and statistical modeling on historical data to forecast future clinical and operational outcomes. This scientific approach provides highly precise insights that help healthcare providers anticipate and prepare for future events.



• Diagnostic analytics

It uses root-cause analysis and data mining to explain why something happened, helping providers identify the specific patterns and correlations behind past events.

• Descriptive analytics

It analyzes historical data to identify past trends and patterns, helping you understand what happened so you can improve future efficiency and patient care.



• Prescriptive analytics

It optimizes healthcare outcomes by recommending specific actions and using "what-if" simulations to compare the potential impact of different decisions.



Benefits of Healthcare Analytics



- Improving decision making



- Addressing patient experience



- Optimising healthcare delivery



- Identifying gaps



- Improving operational efficiency

ENHANCING HEALTHCARE QUALITY THROUGH INFORMATION TECHNOLOGY (IT)

1 PLANNING, DESIGN & IMPLEMENTATION OF HEALTHCARE IT

A structured process of introducing systems like *Electronic Health Records* (EHR) into complex healthcare settings to improve safety and efficiency.



GOAL:

Safe, efficient, patient-centered care supported by well-planned technology.



5 KEY STEPS

(SYSTEM DEVELOPMENT LIFE CYCLE MODEL)



PLANNING — identifies goal and needs.

DESIGN — Align with clinical workflow.



DEVELOP & TEST — Pilot system.

IMPLEMENT — Train staff & launch.



EVALUATE — Monitor outcomes & improve.

QUALITY FLOW

1. DATA QUALITY

- Accurate, complete, and reliable patient data are collected and documented in health information systems.

2. SYSTEM QUALITY

- Secure, efficient, and reliable information technology systems support healthcare operations and data management.

3. DATA ANALYSIS

- Healthcare data are analyzed to identify trends, risks, and patterns that affect patient outcomes.

4. REPORTING AND ACTION

- Findings are reported to healthcare teams to support decision-making and implement quality improvements

2 THE QUALITY SPECTRUM IN INFORMATICS



Ensures healthcare data and systems improve **clinical decision-making**, **patient safety**, and **healthcare outcomes** through effective information management.

GOAL:

- Improve clinical decision-making
- Enhance patient safety
- Support evidence-based practice
- Promote continuous quality improvement
- Improve healthcare system efficiency

TRANSLATION OF EVIDENCE

MEANING

is the process of applying research evidence into clinical nursing practice to improve patient outcomes and healthcare policy.



IMPORTANCE

Bridges the gap between research knowledge and bedside care; essential for advanced practice nurses (APNs) and evidence-based healthcare.

KNOWLEDGE-TO-ACTION CYCLE

1

IDENTIFY A
CLINICAL
PROBLEM

2

SEARCH FOR
THE BEST
EVIDENCE

3

APPRAISE THE
EVIDENCE

4

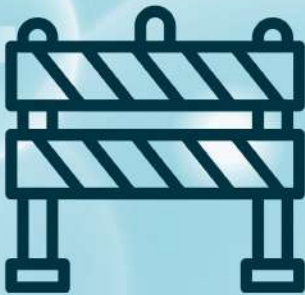
IMPLEMENT
EVIDENCE-BASED
INTERVENTIONS

5

EVALUATE
OUTCOMES



BARRIERS to Translation



Lack of time for clinicians to read and apply research

Limited skills in evaluating or interpreting research

Organizational culture that does not support change

Clinician attitudes and beliefs about research use



Successful Translation

- Consider translation early in research design
- Involve clinical end users (nurses, clinicians) from the start
- Plan for resources to implement and evaluate change
- Get support from senior leaders and stakeholders
- Collect data on whether the implemented change works



INCORPORATING EVIDENCE: USE OF COMPUTER-BASED CLINICAL DECISION SUPPORT SYSTEMS (CDSS) FOR HEALTH PROFESSIONALS

DEFINITION

A Clinical Decision Support System (CDSS) is a health IT system that provides evidence-based guidance to clinicians to improve decision-making and patient care.

Key characteristics

- Pulls patient data (e.g., diagnoses, labs, meds) from EHRs or manual input.
- Uses knowledge bases (rules, guidelines, algorithms, AI) to generate support.
- Provides recommendations, alerts, or interpretations tailored to individual patients.
- Designed to be used at the point of care — when decisions are being made.



MECHANISM OF ACTION



A typical CDSS includes:

1. **Knowledge Base:** Evidence, clinical guidelines, algorithms.
2. **Inference Engine:** Logic that applies the knowledge to patient data
3. **Input Data:** Patient info from EHRs or clinician entries
4. **Output Interface:** Alerts, recommendations, reminders displayed to clinician

EVIDENCE-BASED BENEFITS

Research shows various benefits supported by evidence, especially in improving quality of care and clinician performance:

1. **Improved Clinical Decision-Making:** Provides real-time, evidence-based recommendations for diagnosis, risk assessment, and treatment.
2. **Enhanced Medication Safety:** Reduces prescribing errors (wrong dose, drug interactions) through CPOE integration.
3. **Better Care Processes:** Improves adherence to clinical guidelines (e.g., insulin dosing, hand hygiene), promoting standardized care
4. **Positive Patient Outcomes:** Associated with better glucose control, improved falls & pressure ulcer management, and more accurate emergency triage.



CHALLENGES AND LIMITATIONS



1. **Alert Fatigue:** Excessive system alerts may overwhelm clinicians, leading to ignored or overridden warnings.
2. **Workflow Disruption:** Poor integration with electronic health records (EHR) can interrupt clinical workflow and reduce efficiency.
3. **Cost & Infrastructure Barriers:** Implementation requires financial investment, IT support, maintenance, and continuous staff training — especially difficult in low-resource settings.

CONCLUSION

CDSS strengthens evidence-based practice by improving decision-making, safety, and care consistency. Its success depends on proper integration, training, and alignment with local practice. CDSS supports — but does not replace — professional clinical judgment.

NURSING CURRICULUM *Reform & Healthcare* INFORMATION TECHNOLOGY (HIT)

NURSING CURRICULUM

WHAT IS IT ?

- Updating nursing education to match modern healthcare demands, technology, and patient needs.

IMPORTANCE

- Rapid advancement in medical technology
- Increase in chronic diseases
- Patient safety concerns
- Global health challenges (e.g., pandemics)
- Shift to evidence-based practice

Key Reforms in Nursing

1. Competency-Based Education

- Focus on skills, critical thinking, and clinical judgment
- Emphasizes outcomes, not just classroom hours

2. Evidence-Based Practice (EBP)

- Teaching students to use research in decision-making

3. Interprofessional Education (IPE)

- Collaboration with medicine, pharmacy, and allied health students

4. Simulation & Skills Labs

- High-fidelity mannequins
- Virtual case scenarios
- Safer learning environment before real patients

5. Informatics Integration

- Training in electronic charting



HEALTHCARE INFORMATION TECHNOLOGY (HIT)



- The use of digital systems to store, manage, and share health information.

Major Components of HIT

1. Electronic Health Records (EHRs)

- Digital patient charts
- Reduce errors from illegible handwriting
- Improve continuity of care

2. Telehealth / Telemedicine

- Remote consultations
- Increased access for rural patients

3. Clinical Decision Support Systems (CDSS)

- Alerts for drug interactions
- Reminders for preventive care

4. Health Information Exchange (HIE)

- Sharing patient data across facilities

Why HIT matters in Nursing

- ✓ Improves patient safety
- ✓ Reduces medication errors
- ✓ Enhances communication
- ✓ Increases efficiency
- ✓ Supports data-driven decisions

CONNECTION BETWEEN CURRICULUM REFORM & HIT

MODERN NURSING CURRICULUM NOW INCLUDES:

- Nursing informatics
- Electronic documentation training
- Data privacy & cybersecurity
- Use of simulation technologies
- Exposure to real EHR systems during clinical rotations

TECHNOLOGY INFORMATICS GUIDING EDUCATION REFORM

MODULE 6



INFORMATICS COMPETENCIES FOR FACULTY MEMBERS

Faculty need skills in:

- Basic Computer and information literacy.
- EHR documentation and data management.
- Online teaching platforms (LMS, virtual simulation).
- Data privacy, ethics, and cybersecurity.
- Evidence-based practice using digital database.

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THE DECADE OF HEALTHCARE TECHNOLOGY (2004)

Around 2004, global healthcare accelerated adoption of:

- Electronic Health Records (EHRs)
- Clinical Decision Support Systems
- Telehealth
- Health Information Exchanges

This period marked the shift toward digital, data-driven care systems.



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BIRTH OF THE TIGER INITIATIVE (2005)

The TIGER was launched to:

- Integrate informatics into nursing education
- Improve nurses' IT competencies
- Support safe, technology-enabled patient care



TIGER COLLABORATIVE WORKGROUPS (2007-2008)

Focused on:

- Informatics competencies
- Faculty development
- Staff development
- Leadership and policy
- Usability and standards

These workgroups created frameworks still used in nursing education.

ONCE A TIGER, ALWAYS A TIGER

Promotes lifelong informatics learning and continuous digital competency development among healthcare professionals.

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TIGER INITIATIVE (2009-2014)

During this phase:

- Competency models were refined.
- Global partnerships expanded
- Integration with organizations like Health Information and Management Systems Society (HIMSS) strengthened

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FACULTY SUPPORT

Includes:

- IT training workshops
- Instructional design assistance
- Access to digital tools
- Technical support services

Strong faculty support improves online teaching quality.



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TECHNOLOGIES USED IN ONLINE LEARNING

- Learning Management Systems (e.g., Moodle, Canvas)
- Virtual simulation & VR
- Video conferencing tools
- Digital assessment platforms
- AI-supported tutoring systems



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MOBILE COMPUTING IN HEALTHCARE & EDUCATION

- Smartphones and tablets for EHR access
- Clinical apps for drug references
- Mobile learning modules
- Telehealth via mobile devices

STUDENT SUPPORT

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- Academic advising
- Technical support
- Online library access
- Mental health services
- Peer collaboration tools

Essential for retention and success in online education.

Improves accessibility and real-time clinical decision-making.

COURSE DEVELOPMENT (ONLINE)

Effective online courses include:

- Clear learning outcomes
- Interactive activities
- Multimedia resources
- Evidence-based content
- Aligned assessments

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LEGAL, ETHICAL & COPYRIGHT

Issues Key concerns:

- Patient data privacy (confidentiality)
- Academic integrity
- Copyright compliance
- Cybersecurity risks

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Program Evaluation & Accreditation

Evaluation ensures:

- Learning outcomes are achieved
- Curriculum meets professional standards
- Continuous quality improvement

Standards for Quality Online Education

Organizations like "Quality Matters" provide:

- Course design rubrics
- Peer review systems
- Evidence-based online teaching standards

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Future Trends

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- Artificial Intelligence in education
- Big data analytics
- Virtual & augmented reality
- Personalized adaptive learning
- Global digital collaboration

Effectiveness of Online Education

Research shows online education:

- Can be as effective as face-to-face learning
- Improves flexibility and access
- Requires strong instructional design



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The Connected Age

We live in a digitally interconnected world where:

- Information flows instantly
- Healthcare is data-driven
- Learning happens anytime, anywhere



SOCIAL MEDIA TOOLS IN THE CONNECTED AGE:

Impact on Healthcare Education and Healthcare Practices

HOW SOCIAL MEDIA SHAPES THE HEALTHCARE FIELD

Digital tools like social media have changed how people manage their health by making them more involved in healthcare decisions. With mobile devices, people can access information anytime and connect with other patients and healthcare professionals, helping them learn more and take a more active role in their health.

HOW IT STARTED

1960s–1980s Internet Era

- Foundational global network for data exchange.
- Early access to digital health data; limited interaction.



1991–2004 Web 1.0

- Static web pages accessible through the WWW.
- Patients can access health information online; info dissemination expands.



2004 – Late 2000s Web 2.0

- Interactive platforms, user-generated content, early social networks.
- Healthcare students & professionals share content; communities form online.



Late 2010s - Present Connected Age

- Real-time, mobile, rich multimedia connections; social ecosystems.
- Social media integrated into healthcare education and practice; engagement, collaboration, patient empowerment increase.



From Web 1.0 static sites to Web 2.0 interactive platforms and now the Connected Age, the evolution of social media has shifted healthcare from passive information delivery to active collaboration, continuous learning, and connected care—benefiting both education and clinical practice by empowering patients and professionals alike.

KEY IMPACTS IN HEALTHCARE

Social media enhances patient participation, communication, collaboration, and information availability, all of which contribute to better healthcare. But there are disadvantages as well, like threats to privacy, false information, issues with professionalism, and unequal access. Healthcare workers must be knowledgeable about technology, ethically conscious, and have clear institutional policies in order to minimize risks and increase benefits.



BENEFITS AND CHALLENGES OF SOCIAL MEDIA IN HEALTHCARE

HEALTHCARE EDUCATION

- **Access to Information:** Easy access to medical resources and tutorials.
- **Collaborative Learning:** Online sharing of knowledge with peers and mentors.
- **Interactive Learning:** Use of videos and podcasts for better understanding.
- **Global Networking:** Connection with experts and institutions worldwide.

- **Misinformation:** Students may encounter inaccurate or non-evidence-based content.
- **Digital Literacy Gap:** Not all learners can critically evaluate online resources.
- **Privacy & Confidentiality Risks:** Sharing case studies or patient information can breach ethical standards.

HEALTHCARE PRACTICE

- **Patient Engagement & Education:** Social media helps spread health awareness and patient education.
- **Professional Collaboration:** Healthcare workers share experiences, research, and best practices.
- **Telehealth & Remote Support:** Apps and devices allow monitoring, consultations, and guidance.
- **Rapid Communication:** Provides real-time updates during emergencies like COVID-19.
- **Community Support:** Online groups give peer support and help patients manage their conditions.

- **Patient Privacy Concerns:** Sharing patient information online can break privacy laws like HIPAA.
- **Professionalism Issues:** Posting inappropriate content can harm a healthcare worker's reputation.
- **Information Overload:** Too much or conflicting information can confuse patients and providers.
- **Cybersecurity Threats:** Hackers may steal patient or hospital data.
- **Unequal Access:** Not everyone has good internet or social media access, which can increase inequality.

COMMONLY USED PLATFORMS

- **YouTube/Instagram:** Used for medical education, infographics, and short-form videos.

- **WhatsApp:** Frequently used for professional communication and patient interaction.

- **Facebook/X (Twitter):** Utilized for community building, health campaigns, and networking.

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PLEASE SEE THE QR CODE FOR FURTHER INFORMATION AND REFERENCES

